



Accelerating Design Exploration and Optimization with Surrogate Physics Models

Files and additional resources can be found at https://learn.ntop.com/asme_hackathon/

Background

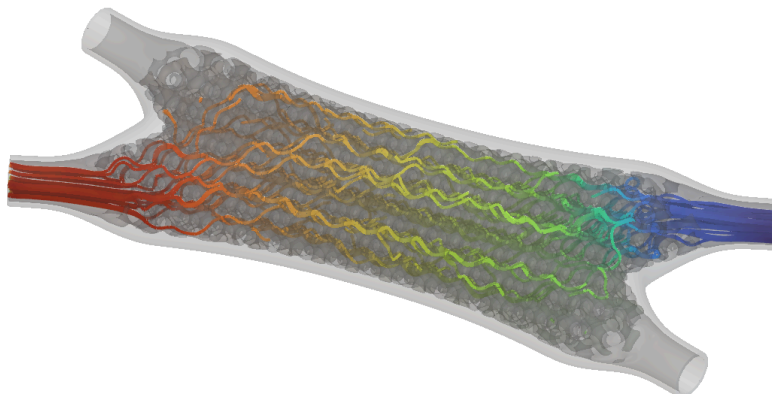
When optimizing a complex part, each iteration takes a long time, both due to geometry generation and simulation compute time. One approach is to sample the design space and use machine learning to train a surrogate physics model from high-quality simulation data. Unlike generic black-box ML models, this surrogate is grounded in the parameter space of interest and reflects the governing physics captured by the simulations. This allows for quicker iteration and trustworthy inverse design approaches.

Problem Statement

Using the data set containing CFD results and mass properties from a heat exchanger model, train a surrogate model capable of predicting pressure drop, average flow velocity, core surface area, and mass given input lattice cell size in the X-direction and Y/Z-direction, and inlet flow velocity. Once your model is trained, use inverse design to specify the optimal lattice cell sizes and flow velocity to maximize the surface area while satisfying the specified constraints for pressure drop, mass, and average flow velocity. The value of a surrogate model is measured in both accuracy and speed, so you will also be evaluated on how long it takes for your model to generate a prediction given input parameters.

Generating Additional Data and Testing Model

We have also provided the nTop file of the parameterized heat exchanger design. It is not required, but you are welcome to generate additional data for training or evaluating your model using nTop Fluids (you can request a [free nTop EDU License here](#)). Your license will include access to nTop Automate, which allows you to run your nTop notebook through a script (example Python file also provided). You are not required to generate additional data, but doing so could help you when training your model for inverse design.





Submission

Your submission should be shared as a zip file or a Public GitHub repo containing the model so it can be run on the judge's computer. The judge should be able to input a set of parameters and receive a prediction for the part's pressure drop, average velocity, surface area, and mass for a given set of input parameters (Cell Size X, Cell Size Y/Z, Inlet Velocity). Your model should also be capable of predicting an optimal set of input parameters given the scoring criteria below using inverse design (input parameters can be specified up to 6 significant figures).

Please also provide a short report detailing your training approach and why you chose the method you did, along with any relevant visualizations of your model performance. This report can be the same as the presentation slides or an additional document.

Data Set and Model Evaluation

The data set, evaluation inputs, and optimization results will be constrained to the following ranges:

- $10\text{mm} < \text{Cell Size X} < 25\text{mm}$
- $10\text{mm} < \text{Cell Size Y/Z} < 25\text{mm}$
- $2500 \text{ mm/s} < \text{Inlet Velocity} < 3500 \text{ mm/s}$

The data set was generated with a "Cell Size" parameter of 0.4mm for the Flow Analysis block and will be the same for evaluating your model.

Notes

- The challenge focuses on developing a methodology for training and implementing surrogate physics models, not on the accuracy of the simulation results.
- The data set was generated using a full factorial experiment design, even though that might not be the most efficient or effective method for training ML models.



Submissions will be scored in the following categories:

Category	Criteria	Score/Metrics
Model Prediction (40%)	How closely does your model predict the pressure drop of a specified cell size compared to simulation results in nTop Fluids, and how long does the prediction take?	• RMSE of model prediction compared to simulation results • The time it takes for your model to predict
Inverse Design (40%)	Determine the optimal lattice cell sizes and inlet velocities that maximize the surface area while meeting the following constraints: • Mass < 125 grams • Pressure Drop < 8000 Pa • Avg Velocity > 520 mm/s²	• Performance of the parameters prescribed by your model
Presentation (20%)	How well do you describe your approach to training a surrogate physics model and how well can your methodology be applied to come complex design problems?	• Clarity of documentation and presentation • Broader application and potential impact of methodology